

## **MODEL QUESTION PAPER**

### **INORGANIC CHEMICAL TECHNOLOGY**

Max.Marks : 75

#### **Part A**

I. Answer all questions in one word or one sentence. Each question carries 1 mark

9 X 1 = 9 Marks

1. Identify the byproducts of caustic soda industry
2. List any two uses of bleaching powder
3. Enlist the raw materials used for hydrochloric acid manufacture
4. Write the reactions in the manufacturing of nitric acid
5. List commonly used nitrogenous fertilizers
6. Define complex fertilizers
7. Name the kind of reaction taking place when compressed ammonia and  $\text{CO}_2$  are reacted
8. List the types of clays
9. List the uses of refractory's

#### **Part B**

II. Answer any EIGHT questions from the following. Each question carries 3 marks

8 X 3 = 24 Marks

1. Sketch and label the diagram of membrane cell
2. List the industrial application of chlorine
3. Identify the sources of raw material for hydrochloric acid manufacture
4. Define catalyst poisoning
5. List the industrial application of ammonia
6. Enumerate the three requirements of phosphatic fertilizers
7. Elucidate the storage of cement
8. Define anti skinning agents.
9. List the types of ceramic products
10. Identify various refractory products

#### **Part C**

Answer ALL questions. Each question carries 7 marks

6 X 7 = 42 Marks

III. Select brine as a raw material for manufacturing of soda ash . Illustrate the manufacturing process

OR

IV. List the properties of lime

V. Describe the concentration process of nitric acid

OR

VI. Apply the principle of DCDA in manufacturing of sulfuric acid with flow diagram

VII. Illustrate the manufacturing process of phosphoric acid

OR

VIII. Explain steam reforming process for the manufacture of hydrogen

IX. Compare the single superphosphate and triplesuper phosphate with respect to raw materials and uses

OR

X. List disadvantages of biurate formation in manufacturing of urea

XI. Outline the importance of mixed fertilizer in agriculture sector.

OR

XII. Describe the raw material preparation steps for the manufacture of titanium dioxide

XIII. Summaries the physical and refractory properties of clay

OR

XIV. Summaries the properties of good refractoriness

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**Course : INORGANIC CHEMICAL TECHNOLOGY**

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**Mark Distribution**

| <b>Module</b> | <b>Hr/Module</b> | <b>Marks / Module</b> | <b>Type of Questions</b> |              |                        |              |                        |              |                        |              |
|---------------|------------------|-----------------------|--------------------------|--------------|------------------------|--------------|------------------------|--------------|------------------------|--------------|
|               |                  |                       | <b>Part A</b>            |              | <b>Part B</b>          |              | <b>Part C</b>          |              | <b>Total</b>           |              |
|               |                  |                       | <b>No of Questions</b>   | <b>Marks</b> | <b>No of Questions</b> | <b>Marks</b> | <b>No of Questions</b> | <b>Marks</b> | <b>No of Questions</b> | <b>Marks</b> |
| 1             | 12               | 22                    | 2                        | 2            | 2                      | 6            | 2                      | 14           | 6                      | 22           |
| 2             | 16               | 39                    | 2                        | 2            | 2                      | 6            | 3                      | 21           | 7                      | 39           |
| 3             | 20               | 50                    | 3                        | 3            | 4                      | 12           | 5                      | 35           | 12                     | 50           |
| 4             | 12               | 22                    | 2                        | 2            | 2                      | 6            | 2                      | 14           | 6                      | 22           |
| Total         | 60               | 123                   | 9                        | 9            | 10                     | 30           | 12                     | 84           | 31                     | 123          |

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**Cognitive Level Mark Distribution**

| <b>Cognitive Level</b> | <b>Marks</b> | <b>% of Marks</b> |
|------------------------|--------------|-------------------|
| Remembering            | 43           | 27                |
| Understanding          | 80           | 50                |
| Applying               |              |                   |
| Analysing              | -            |                   |
| Evaluating             | -            |                   |
| Creating               | -            |                   |
| <b>Total</b>           | <b>123</b>   | <b>100</b>        |

**Question Wise Analysis**

**Course : INORGANIC CHEMICAL TECHNOLOGY**

| <b>Q.No</b> | <b>Module Outcome</b> | <b>Cognitive Level</b> | <b>Marks</b> | <b>Time</b> |
|-------------|-----------------------|------------------------|--------------|-------------|
| I.1         | M 1.01                | R                      | 1            |             |
| I.2         | M 1.04                | R                      | 1            |             |
| I.3         | M 2.02                | R                      | 1            |             |
| I.4         | M 2.03                | U                      | 1            |             |
| I.5         | M 3.01                | R                      | 1            |             |
| I.6         | M 3.02                | R                      | 1            |             |
| I.7         | M 4.02                | R                      | 1            |             |
| I.8         | M 3.04                | R                      | 1            |             |
| I.9         | M 4.04                | R                      | 1            |             |
| II.1        | M 1.01                | U                      | 3            |             |
| II.2        | M 1.01                | R                      | 3            |             |
| II.3        | M 2.02                | R                      | 3            |             |
| II.4        | M 2.03                | U                      | 3            |             |
| II.5        | M 3.01                | R                      | 3            |             |
| II.6        | M 3.02                | R                      | 3            |             |

|       |        |   |   |  |
|-------|--------|---|---|--|
| II.7  | M 3.03 | R | 3 |  |
| II.8  | M 3.04 | U | 3 |  |
| II.9  | M 4.02 | R | 3 |  |
| II.10 | M 4.04 | R | 3 |  |
| III.  | M 1.02 | U | 7 |  |
| IV.   | M 1.03 | R | 7 |  |
| V.    | M 2.03 | U | 7 |  |
| VI.   | M 2.01 | U | 7 |  |
| VII.  | M 2.04 | U | 7 |  |
| VIII. | M 2.01 | U | 7 |  |
| IX.   | M 3.02 | U | 7 |  |
| X.    | M 3.01 | R | 7 |  |
| XI.   | M 3.02 | U | 7 |  |
| XII.  | M 3.04 | U | 7 |  |
| XIII. | M 4.02 | U | 7 |  |
| XIII. | M 4.03 | U | 7 |  |